

ALL OF THE DOCUMENTS PREPARED BY PROSCENIUM ARCHITECTURE + INTERIORS INC. OR ON BEHALF OF PROSCENIUM ARCHITECTURE + INTERIORS INC. IN CONNECTION WITH THE PROJECT ARE INSTRUMENTS OF SERVICE FOR THE EXECUTION OF THE PROJECT. PROSCENIUM ARCHITECTURE + INTERIORS INC. RETAINS THE PROPERTY AND COPYRIGHT IN THESE DOCUMENTS. WHETHER THE PROJECT IS EXECUTED OR NOT, THESE DOCUMENTS MAY NOT BE USED ON ANY OTHER PROJECT WITHOUT PRIOR WRITTEN AGREEMENT OF PROSCENIUM ARCHITECTURE + INTERIORS INC. DO NOT SCALE DRAWINGS.

LANDSCAPE ARCHITECT
GAUTHIER & ASSOCIATES

STRUCTURAL ENGINEER

RJC

MECHANICAL ENGINEER

AME

ELECTRICAL ENGINEER

AES

GEOTECHNICAL ENGINEER

ENVELOPE

RJC

SURVEYOR

CIVIL ENGINEER

ELECTRICAL CONSULTANT



505 Burrard Street, Suite 950, Vancouver, BC V7X 1M4

604.569.6500 | www.aesengr.com

CALGARY | KELOWNA | VANCOUVER | VICTORIA

14	IFC (BALANCE OF BUILDING)	NF	2024-02-22
13	REVISED FOR VALUE ENGINEERING	NF	2024-02-22
12	REVISED FOR COORDINATION	NF	2024-02-22
11	ISSUED FOR DEMOLITION & CONSTRUCTION FOR FOUNDATION TO GRADE	BO	2024-01-28
10	REVISED FOR BP	BO	2024-01-05
9	ISSUED FOR ELECTRICAL ADDENDUM NO.2	NF	2023-10-27
8	ISSUED FOR ELECTRICAL ADDENDUM NO.1	NF	2023-10-20
7	REVISED FOR BP	BO	2023-10-12
6	ISSUED FOR TENDER	BO	2023-10-06
5	ISSUED FOR BUILDING PERMIT	DS	2023-08-09
4	ISSUED FOR 100% DD	DS	2023-06-03
3	ISSUED FOR 50% DD	DS	2023-02-10
2	ISSUED FOR SD REPORT REVISION 1	DS	2022-11-04
1	ISSUED FOR SD REPORT	DS	2022-10-06

NO.	DESCRIPTION	BY	DATE
CURRENT REVISION			

PROJECT ADDRESS

2080 WEST MALL, VANCOUVER
BC

PROJECT

JACK BELL RENEWAL

DRAWING TITLE

ELECTRICAL
SINGLE LINE
DIAGRAM

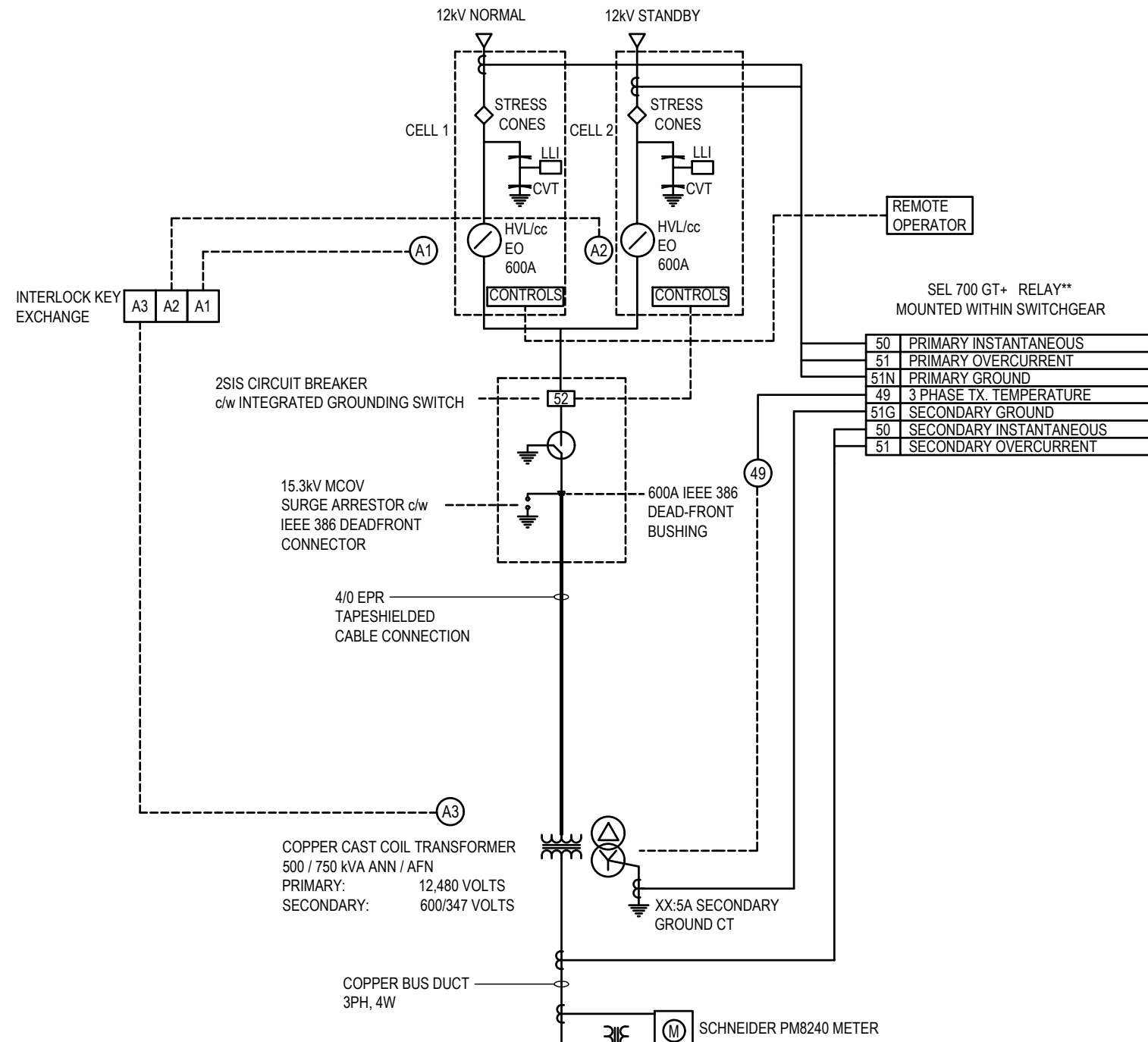
PROJECT NUMBER

0222.0330

DWG. NO.

E250

DATE ISSUE DATE	SCALE AS NOTED	DRAWN DS	CHECKED BOY
--------------------	-------------------	-------------	----------------

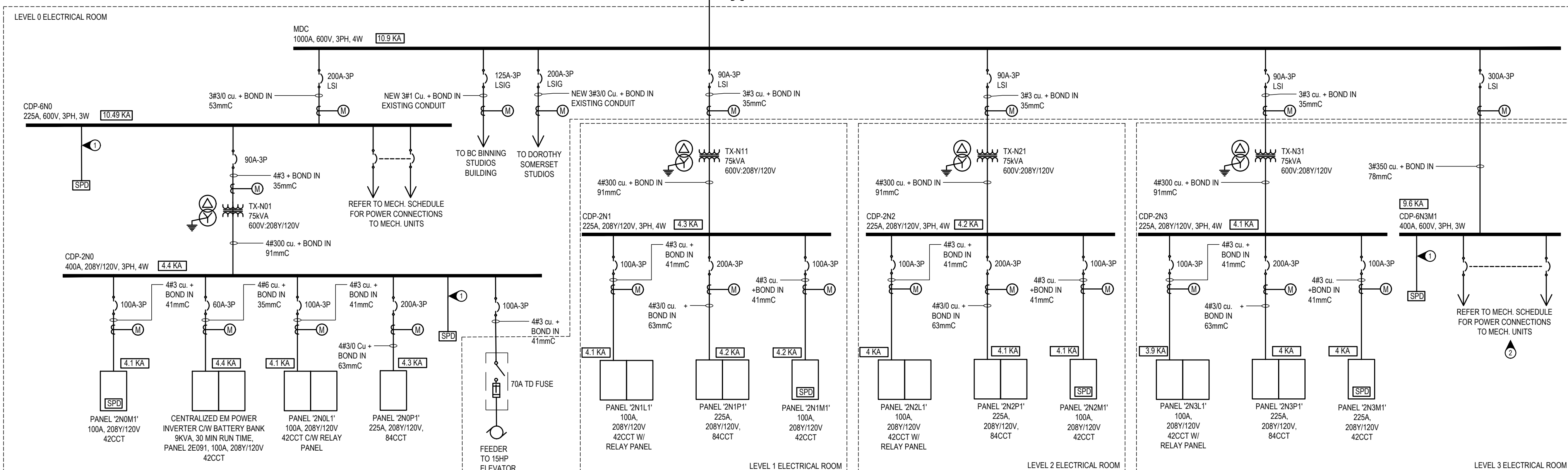


GENERAL NOTES:

- EXISTING UNIT SUBSTATION, TRANSFORMERS, AND ELECTRICAL PANELS TO BE REPLACED WITH NEW UNITS.
- ALLOW FOR REPLACEMENT OF EXISTING PRIMARY AND SECONDARY CONDUCTORS.
- ALL 208Y/120V PANELS SHALL HAVE MINIMUM I_{sc} OF 10KA. ALL 600V PANELS SHALL HAVE MINIMUM I_{sc} OF 22KA. INCREASE PANEL RATING AS SHOWN ON SINGLE LINE DIAGRAM.
- SURGE PROTECTION DEVICE (SPD) SHALL BE INTEGRAL TO THE PANEL THEY ARE SHOWN CONNECTED TO. BREAKER SIZE FOR SPD SHALL BE BASED ON MANUFACTURER RECOMMENDATIONS.
- CONTRACTOR TO SUPPLY, INSTALL, AND COMMISSION REVENUE GRADE METERS (NOTED AS "RM") AND INFORMATION GRADE METERS (NOTES AS "IM") AS SHOWN ON THE DRAWING.

DRAWING KEY NOTES:

- PROVIDE OVER-CURRENT PROTECTION DEVICE FOR SPD AS PER MANUFACTURER RECOMMENDATION (TYPICAL).
- PROVIDE DIGITAL METERS ON BRANCH CCT TO MECHANICAL UNIT CH-1 AND AHU-1 FOR MONITORING PURPOSE.



1 SINGLE LINE DIAGRAM E250 NTS

0222.0330.001 - UBC Jack Bell Electrical Load Calculation				
Load Calculation				
Classroom Area		580	m ²	
Laboratory Area		156	m ²	
Public / Service Area		878	m ²	
Office Area		829	m ²	
Gross Building Area		2,809	m²	
Basic Load Calculation				
	W/m²		m²	Total
Classroom	50		580	28,991.5 W
Laboratory	75		156	11,679.8 W
Public Area	10		878	8,781.7 W
Office Area	50		829	41,425.0 W
Mechanical Load				260,000.0 W
Connected Load				350.9 KW
Items				
	W/m² or W	Demand Factor	m²	Total
First 900 m ² of building area	125	75%	900	84,328.7 W
Remaining area	125	50%	1,909	119,219.8 W
Elevator	25,000	100%		25,000.0 W
Electrical Heating	40,000	75%		30,000.0 W
Existing Building - UBC Binning	112,500	75%		84,375.0 W
Existing Building - Somerset Studio	112,500	75%		84,375.0 W
Demand load:				427.3 KW
Total Demand :				427.3 KW
Power Factor (pf):				0.95
Total Demand:				449.8 KVA

2 ELECTRICAL LOAD CALCULATION E250 NTS

